

CSCI 1101 Introduction to Computer Science



LOGIC CIRCUITS

Computers and Electricity

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Gate

A device that performs a basic operation on electrical signals

Circuits

Gates combined to perform more complicated tasks

Gates

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Six types of gates

- NOT
- AND
- OR
- XOR
- NAND
- NOR

NOT Gate

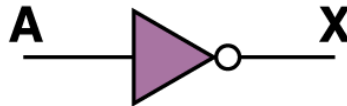
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A NOT gate accepts one input signal (0 or 1) and returns the opposite signal as output

Boolean Expression

$$X = A'$$

Logic Diagram Symbol



Truth Table

A	X
0	1
1	0

AND Gate

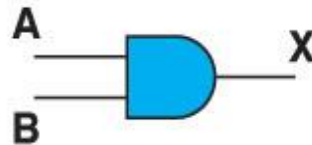
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An AND gate accepts two or more input signals
If all are 1, the output is 1; otherwise, the output is 0

Boolean Expression

$$X = A \cdot B$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate

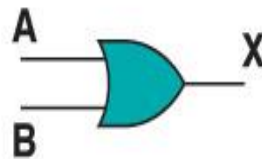
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An OR gate accepts two or more input signals
If all are 0, the output is 0; otherwise, the output is 1

Boolean Expression

$$X = A + B$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

XOR Gate

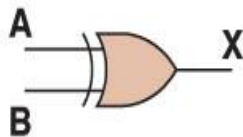
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An XOR gate accepts two or more input signals. If all are the same or number of 1's is even, the output is 0; otherwise, the output is 1.

Boolean Expression

$$X = A \oplus B$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

XOR Gate

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- Note the difference between the XOR gate and the OR gate; they differ only in one input situation
- When both input signals are 1, the OR gate produces a 1 and the XOR produces a 0
- XOR is called the ***exclusive OR***
- OR is the ***inclusive OR***

NAND Gate

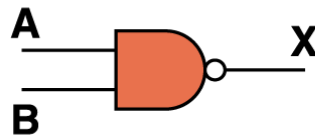


The NAND gate accepts two or more input signals
If all are 1, the output is 0; otherwise, the output is 1

Boolean Expression

$$X = (A \cdot B)'$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

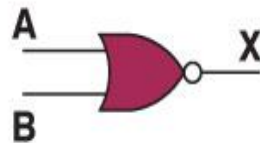
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The NOR gate accepts two or more input signals. If all are 0, the output is 1; otherwise, the output is 0.

Boolean Expression

$$X = (A + B)'$$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

Gates with More Inputs

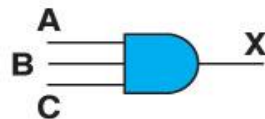
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- Gates can be designed to accept three or more input values
- A three-input **AND** gate, for example, produces an output of **1** only if all input values are **1**

Boolean Expression

$$X = A \cdot B \cdot C$$

Logic Diagram Symbol



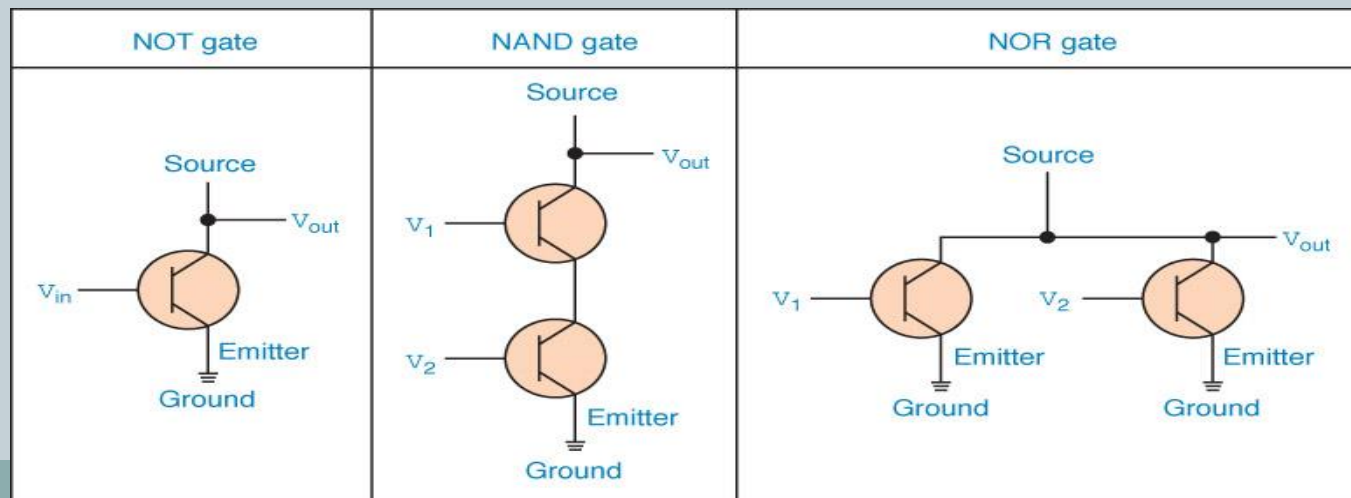
Truth Table

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Constructing Gates

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- Gates are constructed with **transistors**
(**Transistor count:** http://en.wikipedia.org/wiki/Transistor_count)
 - A device that acts either as a wire that conducts electricity or as a resistor that blocks the flow of electricity, depending on the voltage level of an input signal
- The easiest gates to create are the **NOT**, **NAND**, and **NOR** gates
- You will learn more about this in your digital systems course



Circuits

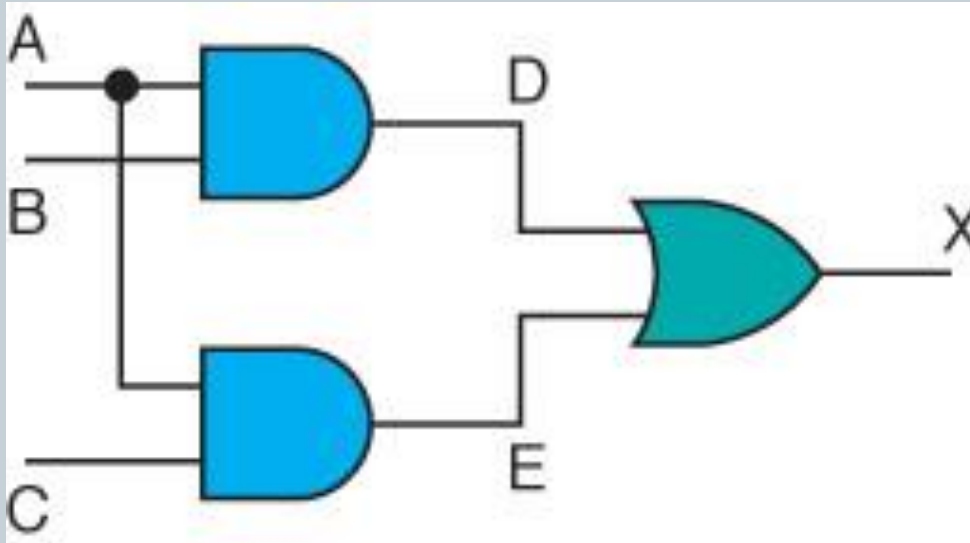
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- **Combinational circuit**
 - The input values explicitly determine the output
- **Sequential circuit**
 - The output is a function of the input values and the existing state of the circuit
 - We describe the circuit operations using
 - Boolean expressions
 - Logic diagrams
 - Truth tables

Combinational Circuits

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Gates are combined into circuits by using the output of one gate as the input for another



Combinational Circuits

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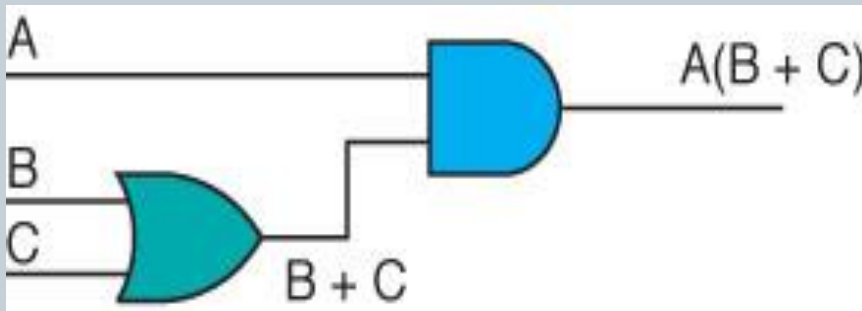
A	B	C	D	E	X
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

- Three inputs require eight rows to describe all possible input combinations
- This circuit is described by the Boolean expression $(AB + AC)$

Combinational Circuits

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Consider the following Boolean expression $A(B + C)$



A	B	C	$B + C$	$A(B + C)$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

Does this truth table look familiar?

Compare it with previous table...

Combinational Circuits

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Circuit equivalence

- Two circuits that produce the same output for identical input
- Boolean algebra allows us to apply provable mathematical principles to help design circuits
- $A(B + C) = AB + AC$ (distributive law) so circuits must be equivalent

